

EA-0017 - AQELYN Engineering Archive

AQELYN - EA-0017 Engineering Archive

IS-017 - AQELYN Threat Detection & Analytics Engine

****Archive ID:**** EA-0017

****Implementation Specification:**** IS-017

****Component:**** AQELYN Threat Detection & Analytics Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0016

****Next Specification:**** IS-018 - AQELYN Automated Response & Orchestration Engine

Document Control

| Field | Value |

|---|---|

| Document | Engineering Archive EA-0017 |

| Specification | IS-017 - AQELYN Threat Detection & Analytics Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0017.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-017 complete; EA-0017 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

|---|---|---|

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine
EA-0015	IS-015	AQELYN Security Operations (SOC) Engine
EA-0016	IS-016	AQELYN Digital Forensics Engine
EA-0017	IS-017	AQELYN Threat Detection & Analytics Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-017 Specification Identity

```
Specification ID: IS-017  
Name: AQELYN Threat Detection & Analytics Engine  
Engineering Archive Target: EA-0017  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-016 - AQELYN Digital Forensics Engine
```

3. Purpose

The AQELYN Threat Detection & Analytics Engine provides real-time detection, behavioral analytics, anomaly identification, attack pattern recognition, and predictive threat analysis across the AQELYN Cyber Security Operating Environment.

It transforms raw telemetry, intelligence, and forensic evidence into actionable threat detections.

It answers:

```
What threats are occurring right now?  
Which activities are anomalous?  
Are multiple events part of the same attack?  
Which identities are behaving abnormally?  
Which assets are under coordinated attack?  
What attacks are likely to occur next?  
How should threats be prioritized?  
What confidence supports each detection?
```

4. Mission

The engine shall provide:

```
Real-time threat detection  
Behavioral analytics  
Anomaly detection  
Correlation analytics  
Attack pattern recognition  
Threat scoring  
Detection confidence calculation  
MITRE ATT&CK mapping  
Predictive analytics  
Threat prioritization  
Detection reporting  
Detection event publication
```

5. Scope

5.1 In Scope

```
Threat detection  
Behavior analytics  
Entity analytics  
Anomaly detection  
Correlation analytics  
Threat scoring  
Detection confidence  
Threat prioritization  
MITRE ATT&CK mapping  
Detection reporting  
Detection event publishing  
Detection dashboards
```

5.2 Out of Scope

```
Malware reverse engineering  
Endpoint antivirus  
Firewall enforcement  
Packet capture  
Network IDS implementation  
Vulnerability scanning
```

6. Dependencies

IS-017 depends on:

```
IS-001 AQELYN Kernel  
IS-002 Universal Object Model  
IS-003 AQELYN Event Bus  
IS-004 AQELYN Evidence Engine
```

- IS-005 AQELYN Knowledge Graph
- IS-006 AQELYN Trust Engine
- IS-007 AQELYN Mission Engine
- IS-008 AQELYN Workflow Engine
- IS-009 AQELYN Policy Engine
- IS-010 AQELYN Compliance & Governance Engine
- IS-011 Identity & Access Governance Engine
- IS-012 Asset & Configuration Governance Engine
- IS-013 Risk Intelligence Engine
- IS-014 Threat Intelligence Fusion Engine
- IS-015 Security Operations Engine
- IS-016 Digital Forensics Engine

7. High-Level Architecture

AQELYN Threat Detection & Analytics Engine

-
- Detection Engine
- Analytics Engine
- Behavioral Analysis Engine
- Anomaly Detection Engine
- Correlation Engine
- Threat Scoring Engine
- MITRE ATT&CK Mapper
- Predictive Analytics Engine
- Detection Dashboard
- Evidence Connector
- Knowledge Graph Connector
- Event Publisher

8. Functional Requirements

FR-017-001 - Threat Detection

The engine shall detect threats using:

- Signature detection
- Behavioral detection
- Heuristic detection
- Rule-based detection
- Correlation-based detection
- Threat intelligence indicators

FR-017-002 - Behavioral Analytics

Analyze behavior for:

- Users
- Devices
- Applications
- Services
- Assets
- Network entities

FR-017-003 - Anomaly Detection

Identify anomalies based on:

- Baseline deviation
- Behavior deviation
- Time deviation
- Access deviation
- Location deviation
- Privilege deviation

FR-017-004 - Correlation Analytics

Correlate:

```
Alerts
Incidents
Evidence
Threat intelligence
Risk intelligence
Mission impact
Identity activity
Asset activity
```

FR-017-005 - Threat Scoring

Calculate threat scores using:

```
Severity
Confidence
Evidence quality
Risk score
Mission criticality
Threat intelligence confidence
```

FR-017-006 - MITRE ATT&CK Mapping

Map detections to:

```
Tactics
Techniques
Sub-techniques
Campaigns
Threat actors
```

FR-017-007 - Predictive Analytics

Estimate:

```
Attack probability
Threat progression
Mission impact
Risk evolution
Likely attacker objectives
```

FR-017-008 - Detection Reporting

Generate:

```
Detection reports
Threat summaries
Analytics reports
Executive dashboards
Detection metrics
```

FR-017-009 - Event Publication

Publish standardized events:

```
threat.detected
threat.scored
anomaly.detected
behavior.analyzed
analytics.completed
prediction.generated
```

9. Non-Functional Requirements

The engine shall provide:

- Real-time analytics
- High scalability
- Evidence-backed detections
- Low latency
- Continuous processing
- Repository stability
- Backward compatibility
- High availability

10. Core Detection Workflow

```
Telemetry Received
↓
Threat Detection
↓
Behavior Analysis
↓
Anomaly Detection
↓
Correlation
↓
Threat Scoring
↓
MITRE ATT&CK Mapping
↓
Prediction
↓
Detection Published
```

11. Internal Component Architecture

The AQELYN Threat Detection & Analytics Engine is implemented as a modular, event-driven subsystem integrated with the AQELYN Kernel, Evidence Engine, Knowledge Graph, Event Bus, and Digital Forensics Engine.

```
AQELYN Threat Detection & Analytics Engine
■
■■■ Detection Engine
■■■ Analytics Engine
■■■ Behavioral Analysis Engine
■■■ Anomaly Detection Engine
■■■ Correlation Engine
■■■ Threat Scoring Engine
■■■ MITRE ATT&CK Mapper
■■■ Predictive Analytics Engine
■■■ Detection Dashboard
■■■ Evidence Connector
■■■ Knowledge Graph Connector
■■■ Risk Intelligence Connector
■■■ Threat Intelligence Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Detection Engine

Responsible for identifying potential threats.

Capabilities:

- Signature detection
- Rule-based detection
- Behavioral detection
- Heuristic detection
- Threat intelligence matching

12.2 Analytics Engine

Processes detection data into operational intelligence.

Functions:

- Detection aggregation
- Confidence calculation
- Detection classification
- Trend analysis
- Statistical analysis

12.3 Behavioral Analysis Engine

Analyzes behavior for:

- Users
- Identities
- Assets
- Endpoints
- Applications
- Services

12.4 Anomaly Detection Engine

Identifies abnormal behavior.

Detection methods:

- Baseline deviation
- Time-series analysis
- Behavior deviation
- Location anomaly
- Access anomaly
- Privilege anomaly

12.5 Correlation Engine

Correlates information from:

- Threat Intelligence
- Risk Intelligence
- Evidence Engine
- SOC Engine
- Digital Forensics Engine
- Knowledge Graph
- Event Bus

12.6 Threat Scoring Engine

Calculates dynamic threat scores.

Factors:

- Severity
- Confidence
- Evidence quality
- Risk score
- Mission impact
- Threat intelligence confidence

12.7 MITRE ATT&CK Mapper

Maps detections to:

- Tactics
- Techniques
- Sub-techniques
- Threat actors
- Campaigns

12.8 Predictive Analytics Engine

Predicts:

- Likely attacker actions
- Threat escalation
- Risk evolution
- Mission disruption
- Future attack probability

12.9 Detection Dashboard

Provides operational visibility.

Displays:

- Active detections
- Threat trends
- Top risks
- MITRE coverage
- Detection confidence
- Mission impact

13. Universal Object Model Extensions

13.1 ThreatDetection

```
ThreatDetection:  
  detection_id  
  source  
  severity  
  confidence  
  timestamp  
  evidence
```

13.2 BehaviorProfile

```
BehaviorProfile:  
  profile_id  
  entity  
  baseline  
  deviations
```

13.3 Anomaly

```
Anomaly:  
  anomaly_id  
  entity  
  confidence  
  severity
```

13.4 ThreatPrediction

```
ThreatPrediction:  
  prediction_id
```

```
probability
expected_attack
confidence
```

14. Knowledge Graph Integration

Relationships:

```
Threat
↓
creates
↓
Detection
Detection
↓
references
↓
Evidence
Detection
↓
affects
↓
Mission
Detection
↓
increases
↓
Risk
Detection
↓
mapped_to
↓
MITRE Technique
```

15. Event Bus Integration

15.1 Detection Events

```
threat.detected
threat.updated
threat.closed
```

15.2 Analytics Events

```
analytics.started
analytics.completed
```

15.3 Behavior Events

```
behavior.profile.updated
behavior.analyzed
```

15.4 Anomaly Events

```
anomaly.detected
anomaly.closed
```

15.5 Prediction Events

```
prediction.generated
prediction.updated
```

16. Evidence Engine Integration

Consumes:

- Evidence quality
- Evidence confidence
- Evidence references
- Artifact metadata

17. Digital Forensics Integration

Consumes:

- Forensic reports
- Timeline analysis
- Memory analysis
- Disk analysis
- Artifact verification

18. Security Operations Integration

Supports:

- Incident creation
- SOC investigations
- Threat hunting
- Alert prioritization
- Response coordination

19. Risk Intelligence Integration

Provides:

- Threat score updates
- Risk correlation
- Risk forecasting
- Mission risk

20. Threat Intelligence Integration

Consumes:

- Indicators of Compromise
- Threat actors
- Campaigns
- Malware families
- Confidence scores

21. Policy Integration

Policies govern:

```
Detection thresholds
Escalation rules
Prediction confidence
Alert publication
Retention
```

22. Public APIs

22.1 Detection API

```
GET /detections
POST /detections
GET /detections/{id}
```

22.2 Analytics API

```
GET /analytics
POST /analytics
```

22.3 Prediction API

```
GET /predictions
POST /predictions
```

22.4 Dashboard API

```
GET /threat/dashboard
GET /threat/metrics
```

23. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
  ■■■ src/
  ■ ■■■ threat_detection/
  ■■■ tests/
  ■ ■■■ threat_detection/
  ■■■ docs/
  ■ ■■■ threat_detection/
  ■■■ api/
  ■ ■■■ threat_detection/
  ■■■ archive/
```

No top-level repository modifications are permitted.

24. Security Architecture

The AQELYN Threat Detection & Analytics Engine is a Tier-1 analytical subsystem responsible for continuously detecting, correlating, prioritizing, and predicting cyber threats across the AQELYN platform.

Every detection shall be:

```
Evidence-backed
Trust-scored
Policy-governed
Explainable
Auditable
Traceable
```

Mission-aware
Risk-aware

24.1 Security Principles

Zero Trust
Defense in Depth
Least Privilege
Continuous Monitoring
Explainable Analytics
Evidence Integrity
Policy Enforcement
Secure by Design

24.2 Authorization Model

Supported operational roles:

SOC Analyst
Threat Hunter
Threat Intelligence Analyst
Senior Analyst
Incident Commander
Mission Owner
Risk Owner
Security Administrator
Automation Service

All privileged analytical operations shall be authorized through the AQELYN Policy Engine.

24.3 Detection Integrity

Detection records shall maintain:

Unique identifier
Detection source
Confidence score
Evidence references
Analytical method
Policy decision references
MITRE mapping
Review history
Audit history

Detection history shall be append-only.

24.4 Analytics Evidence Protection

Analytics evidence shall support:

Immutable evidence references
Evidence confidence
Evidence lineage
Detection reproducibility
Trust scoring
Audit history

The engine shall not overwrite source evidence.

25. Threat Detection Lifecycle

25.1 Detection Lifecycle

```
Telemetry Received
↓
Threat Detected
↓
Correlation
↓
Threat Scored
↓
Evidence Linked
↓
Published
↓
Archived
```

25.2 Behavioral Analytics Lifecycle

```
Behavior Collected
↓
Baseline Updated
↓
Deviation Detected
↓
Anomaly Calculated
↓
Confidence Assigned
```

25.3 Prediction Lifecycle

```
Historical Analysis
↓
Trend Analysis
↓
Threat Forecast
↓
Mission Impact Prediction
↓
Published
```

25.4 Detection Review Lifecycle

```
Detection Created
↓
Analyst Review
↓
Validated
↓
Escalated
↓
Closed
```

26. Continuous Analytics

The engine continuously evaluates:

```
Threat indicators
Behavior changes
Entity relationships
Evidence quality
Mission impact
Enterprise risk
Threat campaigns
Historical patterns
```

27. Performance Requirements

The engine shall support:

- Real-time detection
- Low-latency analytics
- Parallel correlation
- High-volume telemetry
- Scalable detection pipelines
- Continuous prediction

28. Scalability Requirements

The engine shall scale to support:

- Millions of events per hour
- Enterprise-scale telemetry
- Multi-region deployments
- Large Knowledge Graphs
- Large Threat Intelligence datasets
- Hybrid environments

29. Audit Requirements

Every analytical action shall generate immutable audit records.

Audit events include:

- Detection generated
- Threat scored
- Behavior analyzed
- Prediction published
- Analyst validation
- MITRE mapping
- Policy decision

30. Failure Handling

30.1 Detection Failure

- Detection retried
- Failure recorded
- Alert generated

30.2 Analytics Failure

- Analytics suspended
- Retry initiated
- Audit generated

30.3 Correlation Failure

- Detection retained
- Evidence preserved
- Retry scheduled

30.4 Prediction Failure

- Prediction cancelled
- Reason recorded
- Audit updated

31. Testing Strategy

31.1 Unit Testing

Validate:

```
Detection Engine
Analytics Engine
Behavior Analysis Engine
Threat Scoring Engine
MITRE Mapper
Prediction Engine
```

31.2 Integration Testing

Verify interaction with:

```
Kernel
Evidence Engine
Knowledge Graph
Threat Intelligence
Risk Intelligence
SOC Engine
Digital Forensics Engine
Workflow Engine
Policy Engine
```

31.3 System Testing

Validate:

```
Threat detection
Behavior analytics
Anomaly detection
Threat scoring
Prediction generation
Dashboard updates
```

31.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Evidence linkage
Audit logging
Detection integrity
```

31.5 Regression Testing

Verify IS-001 through IS-016 remain unaffected.

32. Acceptance Criteria

IS-017 is complete when:

```
Threat Detection Engine implemented
Behavior Analytics implemented
Anomaly Detection implemented
Threat Scoring implemented
MITRE Mapping implemented
Prediction Engine implemented
Repository unchanged
```

Testing documented

33. Repository Validation

Repository structure remains unchanged.

```
AQELYN/  
■■■ src/threat_detection/  
■■■ tests/threat_detection/  
■■■ docs/threat_detection/  
■■■ api/threat_detection/  
■■■ archive/
```

No top-level repository modifications are permitted.

34. Engineering Summary

IS-017 introduces the AQELYN Threat Detection & Analytics Engine, providing enterprise-scale real-time detection, behavioral analytics, anomaly identification, threat scoring, predictive analytics, and MITRE ATT&CK mapping.

Major capabilities include:

```
Threat Detection  
Behavior Analytics  
Anomaly Detection  
Threat Correlation  
Threat Scoring  
MITRE ATT&CK Mapping  
Predictive Analytics  
Detection Dashboards  
Mission-aware Analytics  
Risk-aware Analytics  
Evidence-backed Detection
```

The engine integrates with the Evidence Engine, Threat Intelligence Fusion Engine, Risk Intelligence Engine, Security Operations Engine, Digital Forensics Engine, and Knowledge Graph while maintaining modularity, repository stability, and backward compatibility.

35. Specification Status

```
Specification ID : IS-017  
Title : AQELYN Threat Detection & Analytics Engine  
Status : COMPLETE  
Engineering Archive : READY FOR GENERATION  
Next Artifact : EA-0017
```

Engineering workflow status:

```
EA-0001 COMPLETE  
EA-0002 COMPLETE  
EA-0003 COMPLETE  
EA-0004 COMPLETE  
EA-0005 COMPLETE  
EA-0006 COMPLETE  
EA-0007 COMPLETE  
EA-0008 COMPLETE  
EA-0009 COMPLETE  
EA-0010 COMPLETE  
EA-0011 COMPLETE  
EA-0012 COMPLETE  
EA-0013 COMPLETE  
EA-0014 COMPLETE
```

EA-0015 COMPLETE
EA-0016 COMPLETE
IS-017 COMPLETE
EA-0017 READY FOR GENERATION

36. EA-0017 Engineering Objective

The objective of IS-017 was to introduce a dedicated Threat Detection & Analytics Engine that enables AQELYN to detect threats, analyze behavior, identify anomalies, correlate signals, score threats, map detections to ATT&CK-style techniques, predict likely threat progression, and publish detection events.

The engine extends AQELYN from operations and forensics into real-time analytical threat detection.

37. EA-0017 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

Threat detection
Detection analytics
Behavioral analysis
Anomaly detection
Signal correlation
Threat scoring
MITRE ATT&CK mapping
Predictive analytics
Detection dashboards
Evidence integration
Knowledge Graph integration
Risk and threat intelligence integration
SOC and forensics integration
Detection event publishing

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

38. Major Engineering Decisions

38.1 Decision 1 - Dedicated Threat Detection & Analytics Engine

Detection and analytics responsibilities are implemented as a standalone engine rather than embedded in SOC, Threat Intelligence, or Digital Forensics.

Rationale:

Clear separation of detection analytics from operational response.
Independent lifecycle and scaling.
Better support for high-volume telemetry and correlation.
Improved explainability of analytics and detection confidence.

38.2 Decision 2 - Evidence-Backed Detections

Every detection shall reference immutable evidence and a confidence state.

Benefits:

Detections become auditable.
Analysts can reproduce detection reasoning.
False positives can be reviewed.

SOC and Risk Intelligence can consume trusted detections.

38.3 Decision 3 - Mission-Aware and Risk-Aware Analytics

Detection prioritization incorporates mission and risk context.

Benefits:

High-impact detections are prioritized.
Threat detection aligns with enterprise objectives.
Mission risk can be updated from detection events.
SOC receives operationally meaningful alerts.

38.4 Decision 4 - Event-Driven Detection Pipeline

Detection, analytics, behavior, anomaly, and prediction events are published through the AQELYN Event Bus.

Examples include:

```
threat.detected
threat.updated
threat.closed
analytics.started
analytics.completed
behavior.profile.updated
behavior.analyzed
anomaly.detected
prediction.generated
```

This maintains loose coupling between AQELYN engines.

38.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
ThreatDetection
BehaviorProfile
Anomaly
ThreatPrediction
```

These extend the Universal Object Model without modifying existing object definitions.

39. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Detection, behavior, anomaly, prediction objects |

| IS-003 Event Bus | Detection, analytics, behavior, anomaly, prediction events |

| IS-004 Evidence Engine | Evidence-backed detections and artifact metadata |

| IS-005 Knowledge Graph | Threat, detection, evidence, mission, risk, MITRE relationships |

| IS-006 Trust Engine | Evidence confidence and detection trust |

| IS-007 Mission Engine | Mission impact and mission-aware detection prioritization |

| IS-008 Workflow Engine | Review, validation, escalation, and closure workflows |

| IS-009 Policy Engine | Detection thresholds, escalation, publication, retention policies |

| IS-013 Risk Intelligence Engine | Threat score updates, risk correlation, forecasting |

| IS-014 Threat Intelligence Engine | Indicators, actors, campaigns, confidence scores |

| IS-015 SOC Engine | Incident creation, threat hunting, alert prioritization |

| IS-016 Digital Forensics Engine | Forensic reports, timelines, artifact verification |

No existing engine required redesign.

40. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/threat_detection/  
■■■ tests/threat_detection/  
■■■ api/threat_detection/  
■■■ docs/threat_detection/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

41. Security Impact Summary

The specification introduces threat-detection-specific security controls:

```
Policy-driven analytical authorization  
Evidence-backed detections  
Append-only detection history  
Explainable analytics  
Detection confidence scoring  
Mission-aware prioritization  
Risk-aware prioritization  
Auditable prediction generation  
Analyst validation workflow
```

No reduction in the security posture of existing components was identified.

42. Capabilities Added

The engine enables AQELYN to support:

```
Threat detection  
Behavior analytics  
Entity analytics  
Anomaly detection  
Correlation analytics  
Threat scoring  
Detection confidence  
Threat prioritization  
MITRE ATT&CK mapping  
Predictive analytics  
Detection reporting  
Detection dashboards  
Detection event publishing
```

43. Risks Identified

| Risk | Mitigation |

|---|---|

| Detection false positives | Confidence scoring, evidence linkage, analyst review |

| Detection false negatives | Multiple detection methods and continuous analytics |

| Analytics overload | Scalable pipelines and prioritization |

| Unexplainable predictions | Evidence and trend references with confidence |

| Correlation errors | Knowledge Graph relationships and validation workflow |

| Excessive alert volume | Threat scoring and SOC prioritization |

| Policy threshold misconfiguration | Policy review and governance workflow |

| Evidence gaps | Evidence Connector and detection retention |

No critical architectural risks were identified that require redesign.

44. Verification Summary

The specification defines verification for:

- Unit testing
- Integration testing
- System testing
- Security testing
- Regression testing

Acceptance criteria cover detection engine, behavior analytics, anomaly detection, threat scoring, MITRE mapping, prediction engine, repository validation, and testing documentation.

45. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

- Modular architecture
- Event-driven communication
- Immutable evidence references
- Traceability
- Explainability
- Security by design
- Repository stability
- Backward compatibility
- Governance-before-archive discipline

46. Dependencies

Required:

- EA-0001 through EA-0016
- IS-001 through IS-016

Enables:

- IS-018 and subsequent response, automation, analytics, and operations-dependent components

47. Completion Record

```

Engineering Archive : EA-0017
Implementation Specification : IS-017
Title : AQELYN Threat Detection & Analytics Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

```

48. Archive Index Update

```

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine

```

49. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0017

Current Status:
EA-0017 COMPLETE

Next Implementation Specification:
IS-018 - AQELYN Automated Response & Orchestration Engine

EA-0017 is completed and archived. The engineering workflow is consistent with the project rule:

```
Implementation Specification -> Engineering Archive -> Continue
```

From this point onward, the next engineering artifact is IS-018.

50. Engineering Archive Publication Standard

EA-0017 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```

EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■■■ examples/

```

```
■ ■■■ example_*.md
■
■ ■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■ ■■■ images/
■
■ ■■■ journal/
■ ■■■ Engineering_Journal.md
■
■ ■■■ manifest/
■ ■■■ manifest.json
■
■ ■■■ MD/
■ ■■■ EA-xxxx.md
■
■ ■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■ ■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■ ■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■ ■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

51. Requirements Matrix

Requirement ID Requirement Evidence in Archive Status			
--- --- --- ---			
FR-017-001 Detect threats Sections 8, 12 Complete			
FR-017-002 Analyze behavior Sections 8, 12, 25 Complete			
FR-017-003 Identify anomalies Sections 8, 12, 25 Complete			
FR-017-004 Correlate detection context Sections 8, 12, 14 Complete			
FR-017-005 Calculate threat scores Sections 8, 12, 24 Complete			
FR-017-006 Map detections to ATT&CK Sections 8, 12, 29 Complete			
FR-017-007 Estimate threat progression Sections 8, 12, 25 Complete			
FR-017-008 Generate detection reporting Sections 8, 12 Complete			
FR-017-009 Publish detection events Sections 8, 15, 38 Complete			
NFR-017-001 Real-time analytics Sections 9, 27 Complete			
NFR-017-002 High scalability Sections 9, 28 Complete			
NFR-017-003 Evidence-backed detections Sections 9, 16, 38 Complete			
NFR-017-004 Low latency Sections 9, 27 Complete			
NFR-017-005 Continuous processing Sections 9, 26 Complete			
NFR-017-006 Repository stability Sections 23, 33, 40 Complete			

52. Traceability Matrix

Source Target Relationship		
--- --- ---		
IS-017 Purpose EA-0017 Objective Defines why the engine exists		

Detection Engine	FR-017-001	Implements threat detection
Behavioral Analysis Engine	FR-017-002	Implements behavioral analytics
Anomaly Detection Engine	FR-017-003	Implements anomaly detection
Correlation Engine	FR-017-004	Implements contextual correlation
Threat Scoring Engine	FR-017-005	Implements threat scoring
MITRE ATT&CK Mapper	FR-017-006	Implements technique mapping
Predictive Analytics Engine	FR-017-007	Implements prediction lifecycle
Detection Dashboard	FR-017-008	Implements reporting and metrics
Event Publisher	FR-017-009	Publishes detection events
Evidence Engine Integration	Evidence-backed detections	References immutable evidence
Digital Forensics Integration	Forensic context	Consumes reports, timelines, artifact verification
SOC Integration	IS-015	Supports incident creation and hunting
Risk Integration	IS-013	Supplies risk correlation and forecasting
Threat Intelligence Integration	IS-014	Supplies indicators, actors, campaigns
Policy Integration	Analytical rules	Controls thresholds, escalation, publication, retention
Repository Validation	Repository Standard	Confirms no top-level redesign

53. Engineering Journal

Journal Entry - EA-0017

EA-0017 was created to archive completion of IS-017 - AQELYN Threat Detection & Analytics Engine.

The archive records the expansion of AQELYN into real-time threat detection and analytics. IS-017 defines the structure needed to detect threats, analyze behavior, identify anomalies, correlate signals, score threats, map detections to ATT&CK-style techniques, predict likely threat progression, publish detection events, and support SOC operations with evidence-backed analytics.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Detection analytics must be modeled separately from SOC operations, threat intelligence, and digital forensics. Threat intelligence supplies indicators and context, digital forensics supplies evidence and timelines, SOC consumes detections operationally, and the Threat Detection & Analytics Engine owns real-time detection, behavioral analytics, correlation, scoring, mapping, and prediction.

Governance Note

EA-0017 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

54. Examples

54.1 Example Threat Detection

```
detection_id: DET-0001
source: threat_detection_engine
```

```
severity: high
confidence: 0.89
timestamp: 2026-07-07T12:00:00Z
evidence:
- evidence://telemetry-1001
- evidence://indicator-match-2001
mitre_mapping:
- T1059
```

54.2 Example Behavior Profile

```
profile_id: BP-1001
entity: ID-0001
baseline:
login_hours: "08:00-18:00"
typical_locations:
- office_network
deviations:
- after_hours_admin_access
- unusual_geo_location
```

54.3 Example Anomaly

```
anomaly_id: ANOM-2001
entity: ASSET-0002
confidence: 0.84
severity: moderate
reason: network behavior deviated from established baseline
```

54.4 Example Detection Event

```
{
  "event_type": "threat.detected",
  "detection_id": "DET-0001",
  "severity": "high",
  "confidence": 0.89,
  "reason": "Behavioral detection correlated with high-confidence threat indicator",
  "source_engine": "aqelyn_threat_detection_analytics_engine"
}
```

55. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0017.md
PDF/EA-0017.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_threat_detection.md
```

56. Final Archive Statement

EA-0017 is the Engineering Archive for IS-017 - AQELYN Threat Detection & Analytics Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

EA-0017 COMPLETE
IS-017 COMPLETE
NEXT: IS-018